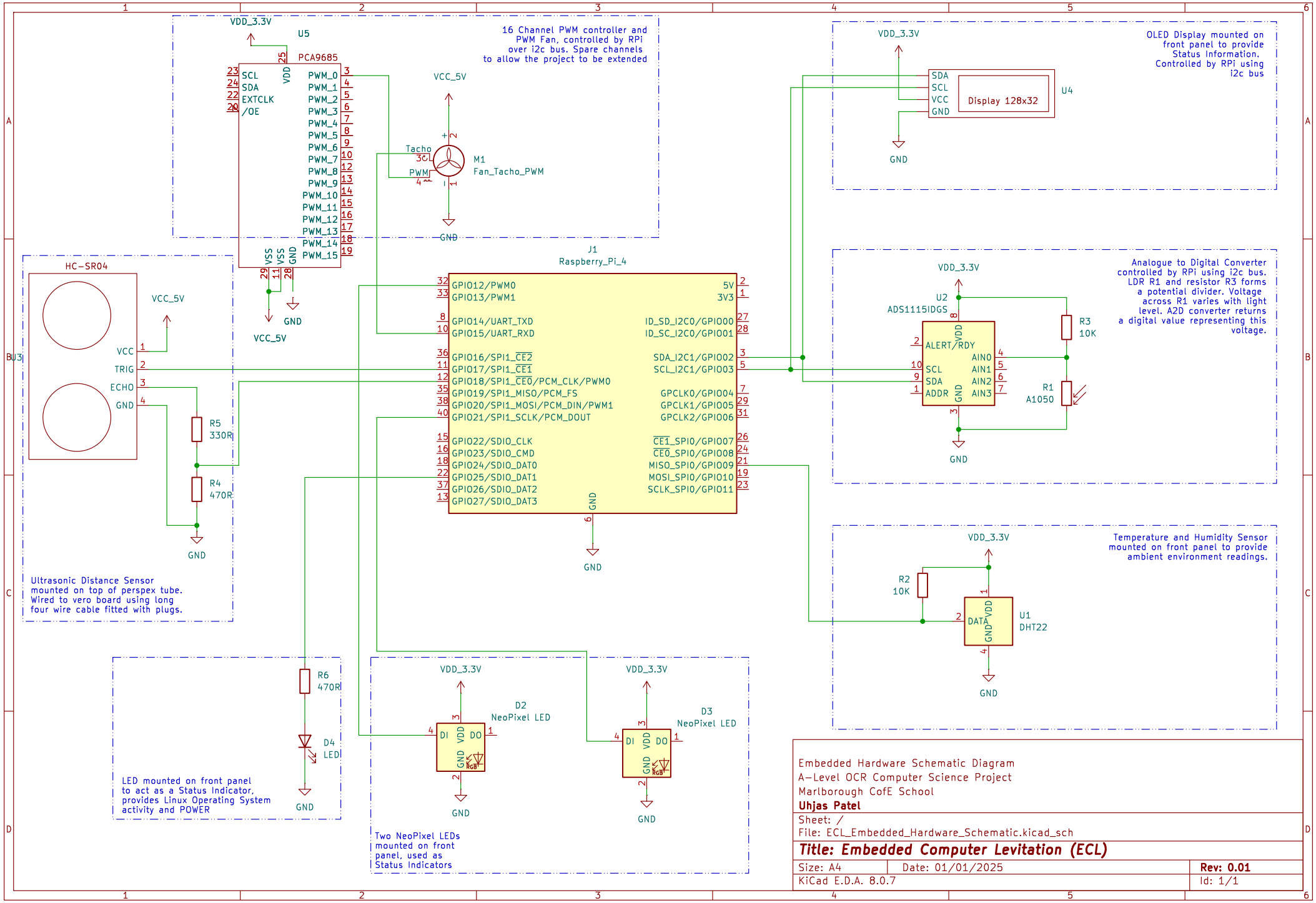

16 Channel PWM controller and PWM Fan, controlled by RPi over i2c bus. Spare channels to allow the project to be extended
U5
PCA9685
VDD_3.3V
VCC_5V
Tacho
PWM
M1
Fan_Tacho_PWM
GND
OLED Display mounted on front panel to provide Status Information. Controlled by RPi using i2c bus
SDA
SCL
VCC
GND
Display 128x32
U4
HC-SR04
VCC
TRIG
ECHO
GND
R5
330R
R4
470R
Ultrasonic Distance Sensor mounted on top of perspex tube. Wired to vero board using long four wire cable fitted with plugs.
J1
Raspberry_Pi_4
GPIO12/PWM0
GPIO13/PWM1
GPIO14/UART_TXD
GPIO15/UART_RXD
GPIO16/SPI1_CE2
GPIO17/SPI1_CE1
GPIO18/SPI1_CE0/PCM_CLK/PWM0
GPIO19/SPI1_MISO/PCM_FS
GPIO20/SPI1_MOSI/PCM_DIN/PWM1
GPIO21/SPI1_SCLK/PCM_DOUT
GPIO22/SDIO_CLK
GPIO23/SDIO_CMD
GPIO24/SDIO_DAT0
GPIO25/SDIO_DAT1
GPIO26/SDIO_DAT2
GPIO27/SDIO_DAT3
5V
3V3
ID_SD_I2C0/GPIO00
ID_SC_I2C0/GPIO01
SDA_I2C1/GPIO02
SCL_I2C1/GPIO03
GPCLK0/GPIO04
GPCLK1/GPIO05
GPCLK2/GPIO06
CE1_SPI0/GPIO07
CE0_SPI0/GPIO08
MISO_SPI0/GPIO09
MOSI_SPI0/GPIO10
SCLK_SPI0/GPIO11
Analogue to Digital Converter controlled by RPi using i2c bus. LDR R1 and resistor R3 forms a potential divider. Voltage across R1 varies with light level. A2D converter returns a digital value representing this voltage.
U2
ADS1115IDGS
ALERT/RDY
SCL
SDA
ADDR
AIN0
AIN1
AIN2
AIN3
R3
10K
R1
A1050
Temperature and Humidity Sensor mounted on front panel to provide ambient environment readings.
R2
10K
U1
DHT22
DATA
R6
470R
D4
LED
LED mounted on front panel to act as a Status Indicator, provides Linux Operating System activity and POWER
D2
NeoPixel LED
D3
NeoPixel LED
DI
DO
Two NeoPixel LEDs mounted on front panel, used as Status Indicators
Embedded Hardware Schematic Diagram
A-Level OCR Computer Science Project
Marlborough CofE School
Uhjas Patel
Sheet: /
File: ECL_Embedded_Hardware_Schematic.kicad_sch
Title: Embedded Computer Levitation (ECL)
Size: A4
Date: 01/01/2025
Rev: 0.01
KiCad E.D.A. 8.0.7
Id: 1/1